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RECURRENCE IN TOPOLOGICAL DYNAMICS AND THE RIEMANN HYPOTHESIS

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1. Introduction

The object of this paper is to point out some surprising connections between two apparently unrelated branches of mathematics. On the one hand we have topological dynamics with its historical origin in the study of particles in motion; on the other hand is the theory of Riemann zeta function and other allied Dirichlet series which arose in the abstract number-theoretic deliberations of Euler, Dirichlet and Riemann.

In its abstract formulation, topological dynamics studies continuous actions of topological groups on topological spaces. In the present work, we allow the additive group of reals (or the additive group of integers) to act on a certain function space H. The Riemann zeta function ζ (as also the other Dirichlet L-functions) may be regarded as a point in the resulting "flow" H. Implicit in [1] was the result that the celebrated Riemann hypothesis on the location of the zeros of zeta is equivalent to strong recurrence (in the sense of Gottschalk and Hedlund in [3]) of zeta regarded as a point in this flow. Analogous results hold for the other Dirichlet L-functions. Here we further elaborate and develop this relationship to show that the generalised Riemann hypothesis holds if and only if all the functions in a naturally defined class of Dirichlet series (viz., those with periodic coefficients) are strongly recurrent points of the flow H. Further, we show that the generalised Riemann hypothesis holds if certain finite subsets of H, each point of which is shown to be strongly recurrent, are jointly strongly recurrent. This makes it interesting to study flows in which separate strong recurrence of points in a finite subset implies joint strong recurrence. By a coherent flow we understand a flow in which this implication is valid. In terms of this definition, the generalized Riemann hypothesis is equivalent to coherence of certain "subspace restrictions" of the flow H. So it is pertinent to ask for sufficient conditions for coherence. To make a beginning in this direction, we show that locally equi-continuous flows are coherent.

2. Flows and other relevant notions from topological dynamics

Throughout this paper G will stand for either the additive group $\mathbf{R}$ of reals or the additive group $\mathbf{Z}$ of integers — as the case may be.

A *flow* is a triplet (X, G, π) where X is a topological space, $G=\mathbf{R}$ or $\mathbf{Z}$, and $\pi\colon X\times G\to X$ is a continuous map satisfying

$$\text{(a)}\quad \pi(x,0)=x, \qquad \text{(b)}\quad \pi(x,t_1+t_2)=\pi(\pi(x,t_1),t_2) \quad \text{for} \quad x\in X,\quad t_1,t_2\in G.$$

In this paper we shall consider flows on metric spaces only. Thus each space X

will be considered equipped with a metric ϱ. If $X_1, X_2, \ldots, X_n$ are metric spaces with metrics $\varrho_1, \ldots, \varrho_n$ respectively, then the product space $X=X_1\times\ldots\times X_n$ will be assigned the metric ϱ given by $\varrho(x,y)=\max_{1\leqq j\leqq n}\varrho_j(x_j,y_j)$. A flow (X,G,π) is said to be a *continuous flow* or a *discrete flow* according as $G=\mathbf{R}$ or $G=\mathbf{Z}$.

If (X,G,π) is a flow and $t\in G$, the map $\pi^t\colon X\to X$ is defined by $\pi^t(x)=\pi(x,t)$, $x\in X$. Clearly each π^t is a homeomorphism of X onto itself. In case of a discrete flow, π^n is clearly the nth iterate of π^1, so that the flow is in this case determined by the surjective homeomorphism π^1. Conversely, each homeomorphism of a space X onto itself induces a discrete flow on X. Thus discrete flow might alternatively be described as a space together with a homeomorphism of the space onto itself.

If (X,G,π) is a flow and Y is a subspace of X, then Y is said to be *invariant* in case whenever $x\in Y$ and $t\in G$, we have $\pi(x,t)\in Y$. In symbols, we have $\pi(Y,G)=Y$. In this case the restriction of π to $Y\times G$, which again we denote by π, determines a flow (Y,G,π). This is called a *subspace restriction* of (X,G,π).

If (X,G,π) is a flow and $A\subseteq X$, let us define the *span of* A, to be denoted by $\mathrm{Sp}\,(A)$, as the minimal invariant set containing A. Notice that we *do not* require the span to be closed. Thus $\mathrm{Sp}\,(A)=\{\pi(x,t)\colon x\in A,\ t\in G\}=\pi(A,G)$. In particular, for any point x in X, $\mathrm{Sp}\,(\{x\})=\pi(x,G)$ is called the *orbit* of x, and is denoted by $O(x)$.

If (X,G,π) is a flow and $h\neq 0, h\in G$, let us define $\pi_h\colon X\times Z\to X$ by $\pi_h(x,n)=\pi(x,nh)$. Clearly (X,Z,π_h) is a discrete flow. We call it the *discrete subflow modulo h*. This is clearly a particular instance of the more general notion of "subgroup restriction" in [4, p. 6].

If for $1\leqq j\leqq n$ (X_j,G,π_j) is a flow (with a common "phase group" G) then their *product flow* (Y,G,π) is given by $Y=X_1\times\ldots\times X_n$ and $\pi=\pi_1\times\ldots\times\pi_n\colon Y\times G\to Y$ where $\pi(x,t)=(\pi_j(x_j,t)\colon 1\leqq j\leqq n)$. In particular, when $X_1=\ldots=X_n=X$, we shall denote the product flow by (X^n,G,π).

If (X_1,G,π_1) and (X_2,G,π_2) are two flows with a common phase group G, and $\varphi\colon X_1\to X_2$ is a continuous map satisfying $\varphi(\pi_1(x,t))=\pi_2(\varphi(x),t)$, $x\in X_1$, $t\in G$, then we say that φ is a *flow homomorphism*. If, in addition, φ is a homeomorphism between X_1 and $\varphi(X_1)$ then we say that φ is an *embedding* of (X_1,G,π_1) in (X_2,G,π_2).

If (X,G,π) is a flow and $x_0\in X$, then we say that x_0 is *periodic* in case there exists $t_0\in G$, $t_0\neq 0$, such that $\pi^{t_0}(x_0)=x_0$. Strong recurrence and almost periodicity are two of the recursion notions which generalise periodicity. We must begin with a few technicalities in order to define them.

If A if a Borel subset of $\mathbf{R}$, the *upper* and *lower asymptotic density* of A, denoted by $\bar{\delta}(A)$ and $\underline{\delta}(A)$, are defined by

$$\bar{\delta}(A)=\limsup_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}I_A(t)\,dt,\quad \underline{\delta}(A)=\liminf_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}I_A(t)\,dt.$$

Here I_A is the indicator function of A: $I_A(t)=1$ if $t\in A$ and $=0$ otherwise. In case $\bar{\delta}(A)=\underline{\delta}(A)$, we call the common value the *asymptotic density* of A, and denote it by $\delta(A)$.

If A is a subset of $\mathbf{Z}$, the *upper* and *lower asymptotic density* of A (discrete

version), denoted respectively by $\bar{d}(A)$ and $\underline{d}(A)$, are defined by

$$\bar{d}(A) = \limsup_{N\to\infty} \frac{\#(A\cap[-N, N])}{2N}, \quad \underline{d}(A) = \liminf_{N\to\infty} \frac{\#(A\cap[-N, N])}{2N}.$$

Here $\#(\cdot)$ stands for the number of elements of $(\cdot)$.

In case $\bar{d}(A)=\underline{d}(A)$, we call the common value the *asymptotic density* (discrete version) of A, and denote it by $d(A)$.

If $A\subseteq\mathbf{R}$ (respectively $A\subseteq Z$) then we say that A is *syndetic* or *relatively dense* in case there exists a compact set $K\subseteq\mathbf{R}$ (respectively a finite set $K\subseteq Z$) such that $A+K=\mathbf{R}$ (respectively $A+K=Z$).

Notice that equivalently a set is syndetic if there is an $l>0$ such that every interval of length l contains an element of the set.

2.1. Definitions. Let (X, G, π) be a continuous (respectively discrete) flow. Let $x_0\in X$. Then x_0 is said to be *strongly recurrent* in case for every $\varepsilon>0$ there exists a Borel set $A\subseteq G$ such that $\bar{\delta}(A)>0$ (respectively $\bar{d}(A)>0$) and $\varrho(\pi^t(x_0), x_0)<\varepsilon$ for all $t\in A$. Here ϱ is the given metric on X. If $h\in G$, $h\neq 0$, then x_0 is said to be *strongly recurrent modulo h* in case x_0, regarded as a point in the discrete subflow (X, Z, π_h), is strongly recurrent.

More generally, a subset Y of X is said to be *jointly strongly recurrent* in case for every $\varepsilon>0$, there exists a Borel set $A\subseteq G$ with $\bar{\delta}(A)>0$ (respectively $\bar{d}(A)>0$) such that $\varrho(\pi^t(y), y)<\varepsilon$ for all $t\in A$ and all $y\in Y$. Further, Y is said to be jointly strongly recurrent modulo h if it is jointly strongly recurrent when regarded as a set in (X, Z, π_h). Notice that strong recurrence is an *orbital property*. That is, a point x_0 in the flow (X, G, π) is strongly recurrent if and only if it is strongly recurrent when regarded as a point in the subspace restriction $(O(x_0), G, \pi)$. More generally, a subset Y of X is jointly strongly recurrent if and only if it is so when regarded as a set in $(\mathrm{Sp}\,(Y), G, \pi)$.

The following proposition is a special case of the "inheritance theorem" first proved by Gottschalk and Hedlund in [3] and later given in an improved form in [4].

2.2. Proposition (inheritance theorem for strong recurrence). *Let (X, G, π) be a flow, $x_0\in X$, $h\in G$, $h\neq 0$. Then x_0 is strongly recurrent if and only if x_0 is strongly recurrent modulo h.*

The next two propositions are important for application but their proofs are trivial. So we omit the proof.

2.3. Proposition. *Let (X_1, G, π_1) and (X_2, G, π_2) be flows with a common phase group and let φ be a flow-homomorphism between them. If a point $x_0\in X_1$ is strongly recurrent then so is $\varphi(x_0)\in X_2$.*

2.4. Proposition. *Let (X, G, π) be a flow and let $Y=\{y_1, \ldots, y_n\}$ be a finite subset of X. Then Y is jointly strongly recurrent if and only if the n-tuple $(y_1, \ldots, y_n)$, regarded as a point in the product-flow (X^n, G, π), is strongly recurrent.*

Propositions 2.2 and 2.4 together yield:

2.5. PROPOSITION. *If (X, G, π) is a flow and Y is a finite subset of X, then for each fixed $h \in G$, $h \neq 0$, Y is jointly strongly recurrent if and only if it is jointly strongly recurrent modulo h.*

PROOF. In view of 2.4, this follows on applying 2.2 to the product-flow (X^n, G, π) with $n = \#(Y)$.

Similarly, 2.3 and 2.4 imply:

2.6. PROPOSITION. *Let $Y = \{y_1, \ldots, y_n\}$ be a finite subset of the flow (X, G, π). Let φ be a flow-homomorphism of the product flow (X^n, G, π) into another flow. If Y is jointly strongly recurrent then $\varphi(y_1, \ldots, y_n)$ is strongly recurrent.*

The chief utility of Propositions 2.2 and 2.5 lie in the fact that they allow us to move back and forth between continuous and discrete flows. Because of them it often suffices to prove results regarding strong recurrence and joint strong recurrence for discrete flows only. This technique will be exploited in the concluding section. Here we use Proposition 2.5 to prove a result which will be used in Section 4:

2.7. PROPOSITION. *Let (X, G, π) be a flow. Let $A = \{x_1, \ldots, x_m\}$ be a jointly strongly recurrent subset of X. Let $B = \{y_1, \ldots, y_n\}$ be a finite subset of X such that all points in B are periodic (not necessarily with the same periods). Then $A \cup B$ is jointly strongly recurrent.*

PROOF. It clearly suffices to prove only the case $n = 1$, since the result then follows by induction on n. Accordingly, let $n = 1$. Let $h \in G$, $h \neq 0$ be a period of y_1. By Proposition 2.5, A is jointly strongly recurrent modulo h. But the orbit of y_1 modulo h is a singleton. Hence trivially $A \cup \{y_1\}$ is jointly strongly recurrent modulo h. Another appeal to Proposition 2.5 completes the proof.

A parallel notion of recursiveness is that of almost periodicity. A point x_0 in a flow (X, G, π) is said to be *almost periodic* in case for every $\varepsilon > 0$ there exists a syndetic set $A \subseteq G$ such that for all $t \in A$, $\varrho(\pi^t(x_0), x_0) < \varepsilon$. Here, as before, ϱ is the given metric on X. As with strong recurrence, we could proceed to define joint almost periodicity and almost periodicity modulo h. Analogues of Propositions 2.2—2.7 hold for almost periodicity also.

Clearly a syndetic subset of G has positive upper asymptotic density. Therefore almost periodicity implies strong recurrence. We shall have a brief occasion to return to almost periodicity in the concluding section.

Most of the notions and results discussed in this section are standard. The author failed to locate Proposition 2.7 in the literature. Clearly analogues of this proposition hold for all recursion notions for which the Gottschelk—Hedlund inheritance theorem is valid. So it must be familiar to experts in the field.

3. Strong recurrence in the flow H and the Riemann hypotheses

3.1. *The space H.* Throughout this paper Ω will stand for the open strip consisting of complex numbers z such that $\frac{1}{2}<\mathrm{Re}(z)<1$. Here $\mathrm{Re}(z)$ is the real part of the complex number z. $H=H(\Omega)$ will stand for the space of all analytic functions on Ω, with the topology of uniform convergence on compacta (u.c.c.).

Let K_n, $n\geqq 1$ be a sequence of compact subsets of Ω which increase to Ω. We also assume that for each $n\geqq 1$, K_n is contained in the interior of K_{n+1}. For $f, g\in H$, we define

$$\varrho(f, g) = \sup_{n\geqq 1} \min\left(\frac{1}{n}, \sup_{z\in K_n} |f(z)-g(z)|\right).$$

As is easy to check, ϱ is a metric on H which induces the u.c.c. topology. Of course, ϱ depends on the choice of the sequence K_n. We shall think of H as equipped with this metric.

3.2. *The flow* $(H, \mathbf{R}, \sigma)$. We now allow the real line to act on H by vertical shift. More precisely, we define $\sigma\colon H\times\mathbf{R}\to H$ by $\sigma(f, t)=g$ where $g(z)=f(z+it)$, $z\in\Omega$. Clearly this makes $(H, \mathbf{R}, \sigma)$ a continuous flow.

3.3. *Dirichlet series, Dirichlet L-functions and the Riemann hypotheses.* A *Dirichlet series* is a series of the type $\sum_{n=1}^{\infty} a_n n^{-z}$, where a_n is a sequence of complex numbers (called the sequence of coefficients of the series) and z is a complex variable. Of particular importance in the theory is the series $\zeta(z)=\sum_{n=1}^{\infty} n^{-z}$ which converges for $\mathrm{Re}(z)>1$, and defines an analytic function there. As is well known, the *zeta function* $\zeta(z)$ may be analytically continued to the entire complex plane except for a singularity at $z=1$. Thus zeta is analytic on Ω, and we may (and do) regard ζ as a point in H.

An *arithmetic function* is a function f from the natural numbers into the complex plane. We often write f_n for $f(n)$ and thus display f as a sequence $\{f_n\}$. An arithmetic function f is said to be *completely multiplicative* in case $f(mn)=f(m)f(n)$ for $m, n\geqq 1$ and $f(1)=1$. An arithmetic function f is said to be *periodic* in case there exists $k\geqq 1$ such that $f(n+k)=f(n)$ for all $n\geqq 1$. In this case k is said to be a period of f, and we say that f is periodic modulo k.

A *Dirichlet character modulo k* is by definition an arithmetic function non-vanishing only for integers co-prime to k which is both completely multiplicative and periodic. It is well known that given any $k\geqq 1$ there are only finitely many Dirichlet characters modulo k. Indeed, the total number of Dirichlet characters modulo k equals the number of integers in the interval $[1, k]$ which are relatively prime to k.

Corresponding to each Dirichlet character χ, one defines the *Dirichlet L-function* $L(., \chi)$ by the series $L(z, \chi)=\sum_{n=1}^{\infty} \chi(n)n^{-z}$. These are natural generalisations of the zeta function. Indeed $\zeta=L(., \chi_0)$ where χ_0 is the unique Dirichlet character

modulo 1. Like zeta, all Dirichlet L-functions admit analytic continuation to the entire plane except possibly at $z=1$. Accordingly we regard the functions $L(.,\chi)$ as points in H.

The Riemann zeta function vanishes at the points $-2, -4, \ldots$ the so called trivial zeros. The classical *Riemann hypothesis* conjectures that all the nontrivial zeros of zeta lie on the line $\operatorname{Re}(z)=\frac{1}{2}$. Likewise, the *generalised Riemann hypothesis* (GRH) for Dirichlet L-functions conjectures that for each character χ, the nontrivial zeros of $L(.,\chi)$ lie on the critical line $\operatorname{Re}(z)=\frac{1}{2}$. In view of the functional equation and the Euler product for $L(.,\chi)$, an equivalent formulation is: $L(z,\chi)\neq 0$ for $z\in\Omega$. That is, $\frac{1}{L(.,\chi)}\in H$. This is the form in which we shall consider these celebrated hypotheses.

3.4. *The relationship between the Riemann hypotheses and strong recurrence.* We begin with the following theorem from [1].

3.5. PROPOSITION. *Let $k\geqq 1$, and let χ_j, $1\leqq j\leqq n$, be distinct Dirichlet characters modulo k. Let f_j, $1\leqq j\leqq n$, be members of H such that $\frac{1}{f_j}$ also belong to H. Then for every $\varepsilon>0$, the set A of all $t\in\mathbf{R}$ such that*

(i) $\sup\limits_{1\leqq j\leqq n} \varrho\big(\sigma^t(L(.,\chi_j)), f_j\big) < \varepsilon$

has positive upper asymptotic density $\bar{\delta}(A)>0$.

Conversely, if f_j, $1\leqq j\leqq n$, are such that for all $\varepsilon>0$ the set A of all $t\in\mathbf{R}$ satisfying (i) *satisfies $\bar{\delta}(A)>0$ then $\frac{1}{f_j}\in H$ for $1\leqq j\leqq n$.*

PROOF. The direct part is a reformulation of Theorem 3.1 in [1]. The converse may be proved following the arguments in the proof of the 'if' part of Theorem 3.7 in [1].

The particular case $n=1$ of Proposition 3.5 is so important that we state it separately as:

3.6. COROLLARY. *Let χ be a Dirichlet character and let $f\in H$. For $\varepsilon>0$, let A_ε denote the set of all real numbers t such that $\varrho\big(\sigma^t(L(.,\chi)), f\big)<\varepsilon$. Then $\frac{1}{f}\in H$ if and only if $\bar{\delta}(A_\varepsilon)>0$ for every $\varepsilon>0$.*

Recall that in the above ϱ is the given metric on H which induces the u.c.c. topology.

Substituting $L(.,\chi)$ for f in Corollary 3.6, we immediately deduce:

3.7. THEOREM. *Let χ be a Dirichlet character. Then the Riemann hypothesis holds for $L(.,\chi)$ if and only if $L(.,\chi)$ is a strongly recurrent point of the flow $(H, \mathbf{R}, \sigma)$.*

Notice that in view of Proposition 2.2, Theorem 3.7 might equivalently be stated in terms of strong recurrence in discrete subflows (H, Z, σ_h), $h \neq 0$.

3.8. REMARKS. For each $k \geqq 1$, there is a unique Dirichlet character χ_0 modulo k defined by $\chi_0(n)=1$ if $(n, k)=1$, $\chi_0(n)=0$ otherwise. χ_0 is called the principal character modulo k; the other characters are called non-principal. The non-trivial zeros of $L(., \chi_0)$ coincide with the nontrivial zeros of zeta. Accordingly, the Riemann hypothesis for $L(., \chi_0)$ is equivalent to the classical Riemann hypothesis. In [2] H. Bohr essentially proved Theorem 3.7 for non-principal characters. His proof depends on the fact that for non-principal characters χ, the Riemann hypothesis is equivalent to convergence of the series $\sum \chi(p)p^{-z}$, $z \in \Omega$, as p runs over the primes. Therefore Bohr's proof does not extend to the principal character. After minor modifications, the main theorem in [2] may be stated as follows. Let $f \in H$ be given by a *convergent* Dirichlet series $\sum_p a_p p^{-z}$, $z \in \Omega$, where p runs over the primes. We also assume that

$$\limsup_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |f(a+it)|^2 \, dt < \infty \quad \text{for} \quad \frac{1}{2} < a < 1.$$

Then f is a strongly recurrent point of the flow $(H, \mathbf{R}, \sigma)$. Bohr deduces the non-principal case of Theorem 3.7 above from this result.

4. Joint strong recurrence and the Riemann hypotheses

Substituting $L(., \chi_j)$ for f_j in Proposition 3.5, we obtain:

4.1. THEOREM. *Let $\chi_1, \ldots, \chi_n$ be Dirichlet characters to a common modulus. Then the following are equivalent:*

(a) *Riemann hypothesis holds for $L(., \chi_j)$, $1 \leqq j \leqq n$.*

(b) *The set $\{L(., \chi_j)\colon 1 \leqq j \leqq n\}$ is a jointly strongly recurrent subset of $(H, \mathbf{R}, \sigma)$.*

For integers h, k, $1 \leqq h \leqq k$, let $\zeta_{h,k} \in H$ be defined by $\zeta_{h,k}(z) = \sum_{n=0}^{\infty} (nk+h)^{-z}$, $\operatorname{Re}(z) > 1$, and hence by analytic continuation. For $k \geqq 1$, let $h_1, \ldots, h_n$ be the integers in $[1, k]$ which are relatively prime to k, and let $\chi_1, \ldots, \chi_n$ be the Dirichlet characters modulo k. The map $\varphi\colon H^n \to H^n$ which sends $(f_1, \ldots, f_n)$ to $(g_1, \ldots, g_n)$ given by $g_j = \sum_{i=1}^{n} \chi_j(h_i) f_i$, $1 \leqq j \leqq n$, is clearly a flow homomorphism of the product flow $(H^n, \mathbf{R}, \sigma)$ onto itself. Its inverse φ^{-1} is given by $f_j = \frac{1}{n} \sum_{i=1}^{n} \overline{\chi_i(h_j)} g_i$ and is again a flow homomorphism. Therefore, by Proposition 2.3, $(f_1, \ldots, f_n)$ is a strongly recurrent point of $(H^n, \mathbf{R}, \sigma)$ if and only if $\varphi(f_1, \ldots, f_n)$ is. In particular, since

$$(L(., \chi_1), \ldots, L(., \chi_n)) = \varphi(\zeta_{h_1,k}, \ldots, \zeta_{h_n,k}),$$

it follows from Proposition 2.4 that $\{L(., \chi_j)\colon 1 \leqq j \leqq n\}$ is jointly strongly recurrent if and only if $\{\zeta_{h,k}\colon 1 \leqq h \leqq k, (h, k)=1\}$ is jointly strongly recurrent. Therefore Theorem 4.1 may be rewritten as:

4.2. THEOREM. *Let* $\chi_1, \ldots, \chi_n$ *be an enumeration of all the Dirichlet characters modulo a fixed integer* $k \geqq 1$. *Then the following are equivalent:*

(a) *Riemann hypothesis holds for* $L(., \chi_j)$, $1 \leqq j \leqq n$.

(b) *The set* $\{\zeta_{h,k}: 1 \leqq h \leqq k, (h, k)=1\}$ *is a jointly strongly recurrent subset of* $(H, \mathbf{R}, \sigma)$.

4.3. LEMMA. *Let* $\chi_1, \ldots, \chi_n$ *be distinct Dirichlet characters to a fixed modulus. Let* $g=\sum_{j=1}^{n} a_j L(., \chi_j)$ *where at least two of the complex numbers* a_j *are non-zero. Then for any arbitrary* $f \in H$, *and for any* $\varepsilon>0$, *the set* A_ε *of all real numbers* t *such that* $\varrho(\sigma^t(g), f)<\varepsilon$ *satisfies* $\bar{\delta}(A_\varepsilon)>0$.

PROOF. In view of Proposition 3.5, it suffices to show that the set $\left\{\sum_{j=1}^{n} a_j f_j: f_j \in H \text{ and } \frac{1}{f_j} \in H\right\}$ is dense in H. Let S be the set of all $f \in H$ such that $\frac{1}{f} \in H$. Since S is closed under multiplication by non-zero complex numbers and since the closure of S contains the constant function 0, it is enough to show that $S+S=\{f_1+f_2: f_1, f_2 \in S\}$ is dense in H. But if $g \in H$ is bounded then there is a complex $\alpha \neq 0$ such that $g(z) \neq \alpha$ for $z \in \Omega$. Hence, putting $f_1(z)=g(z)-\alpha, f_2(z) \equiv \alpha$, we get $g=f_1+f_2, f_1, f_2 \in S$. Thus $S+S$ contains the set of all bounded members of H. Since the latter set is dense in H, we are done.

We have the representation $\zeta_{h,k}=\frac{1}{n} \sum_{i=1}^{n} \overline{\chi_j(h)} L(., \chi_j)$, for $(h, k)=1$. Also, if $k \geqq 3$ then $n \geqq 2$ so that at least two of the coefficients in this formula are non-zero. Hence from the above lemma we deduce:

4.4. PROPOSITION. *Let* $1 \leqq h \leqq k$, $k \geqq 3$, $(h, k)=1$. *Then for any* $f \in H$, *for any* $\varepsilon>0$, *the set* A_ε *of all real numbers* t *such that* $\varrho(\sigma^t(\zeta_{h,k}), f)<\varepsilon$ *satisfies* $\bar{\delta}(A_\varepsilon)>0$.

In particular, substituting $\zeta_{h,k}$ for f in the above, we deduce.

4.5. THEOREM. *Let* $1 \leqq h \leqq k$, $k \geqq 3$, $(h, k)=1$. *Then* $\zeta_{h,k}$ *is a strongly recurrent point of* $(H, \mathbf{R}, \sigma)$.

4.6. DEFINITION. For $k \geqq 1$, let D_k be the span of the set $\{\zeta_{h,k}: 1 \leqq h \leqq k, (h, k)=1\}$. Thus D_k is the set of all points $\sigma^t(\zeta_{h,k})$ with $1 \leqq h \leqq k$, $(h, k)=1$, $t \in \mathbf{R}$. $(D_k, \mathbf{R}, \sigma)$ is a subspace restriction of $(H, \mathbf{R}, \sigma)$.

Since for each $t \in \mathbf{R}$, σ^t is a flow homomorphism of $(H, \mathbf{R}, \sigma)$ into itself, Theorem 4.5 implies that each point in $(D_k, \mathbf{R}, \sigma)$ is strongly recurrent for $k \geqq 3$. If the "generating set" $\{\zeta_{h,k}: 1 \leqq h \leqq k, (h, k)=1\}$ were jointly strongly recurrent, it would likewise follow that any finite subset of $(D_k, \mathbf{R}, \sigma)$ is jointly strongly recurrent. So it is natural to enquire when strong recurrence of the points in a finite set implies joint strong recurrence of the set. This leads us to the following:

4.7. DEFINITION. A flow (X, G, π) will be said to be *coherent* in case whenever A is a finite subset of X such that each point of A is strongly recurrent, it follows that A is jointly strongly recurrent.

In view of the discussion preceding this definition, we have:

4.8. THEOREM. *Let $k \geqq 3$ be an integer. Then the following are equivalent:*
(a) *Riemann hypothesis holds for $L(.,\chi)$ for all Dirichlet characters χ modulo k.*
(b) *The flow $(D_k, \mathbf{R}, \sigma)$ is coherent.*

In the fifth and concluding section, we shall consider the notion of coherence in some details. Here we proceed to obtain yet another equivalent formulation of the Riemann hypotheses.

A *Dirichlet polynomial* is a function of the form $z \to \sum_{n=1}^{N} a_n n^{-z}$. These are entire functions and therefore may be regarded as points in $(H, \mathbf{R}, \sigma)$.

4.9. LEMMA. *Let A be a finite subset of $(H, \mathbf{R}, \sigma)$ which is jointly strongly recurrent. Let B be a finite set of Dirichlet polynomials. Then $A \cup B$ is jointly strongly recurrent.*

PROOF. Let $A=\{f_1, \ldots, f_n\}$, $B=\{g_1, \ldots, g_m\}$. We may write $g_j(z)=\sum_{r=1}^{N} a_{rj} r^{-z}$. That is, $g_j=\sum_{r=1}^{N} a_{rj} h_r$, $1 \leqq j \leqq m$ where h_r is given by $h_r(z)=r^{-z}$, $z \in \Omega$. Each h_r is clearly a periodic point of $(H, \mathbf{R}, \sigma)$. Therefore by Proposition 2.7, $\{f_1, \ldots, f_n, h_1, h_2, \ldots, h_N\}$ is jointly strongly recurrent. That is, by Proposition 2.4, $(f_1, \ldots, f_n, h_1, \ldots, h_N)$ is a strongly recurrent point of the product flow $(H^{n+N}, \mathbf{R}, \sigma)$. Also, the map from H^{n+N} into H^{n+m} which sends $(\theta_1, \ldots, \theta_{n+N})$ into $(\theta_1^*, \ldots, \theta_{n+m}^*)$, with $\theta_j^*=\theta_j$ if $1 \leqq j \leqq n$ and $\theta_{n+j}^* = \sum_{r=1}^{N} a_{rj} \theta_{n+r}$ for $1 \leqq j \leqq m$, is a flow homomorphism of $(H^{n+N}, \mathbf{R}, \sigma)$ into $(H^{n+m}, \mathbf{R}, \sigma)$ which sends the strongly recurrent point $(f_1, \ldots, f_n, h_1, \ldots, h_N)$ into the point $(f_1, \ldots, f_n, g_1, \ldots, g_m)$. Therefore by Proposition 2.3,

$$(f_1, \ldots, f_n, g_1, \ldots, g_m)$$

is strongly recurrent. Hence, by Proposition 2.4, $A \cup B=\{f_1, \ldots, f_n, g_1, \ldots, g_m\}$ is jointly strongly recurrent.

4.10. PROPOSITION. *Let A be a finite subset of $(H, \mathbf{R}, \sigma)$ which is jointly strongly recurrent. Let B be a finite subset of $(H, \mathbf{R}, \sigma)$ such that each point in B is given by a Dirichlet series which converges uniformly over Ω. Then $A \cup B$ is jointly strongly recurrent.*

PROOF. Let $A=\{f_1, \ldots, f_n\}$, $B=\{g_1, \ldots, g_m\}$. Let $\varepsilon>0$ be arbitrary and let us choose $\eta>0$ and a compact set $K \subset \Omega$ such that whenever $h_1, h_2 \in H$ satisfies $\sup_{z \in K} |h_1(z)-h_2(z)|<\eta$, it follows that $\varrho(h_1, h_2)<\frac{\varepsilon}{3}$.

By assumption, there exists Dirichlet polynomials $g_1^*, \ldots, g_m^*$ such that $\sup_{z \in \Omega} |g_j(z)-g_j^*(z)|<\eta$. Hence, by choice of η, we have $\varrho(g_j, g_j^*)<\frac{\varepsilon}{3}$ and $\varrho(\sigma^t(g_j), \sigma^t(g_j^*))<\frac{\varepsilon}{3}$ for all real t.

Now, by the preceding lemma, there exists a set U of reals with $\bar{\delta}(U)>0$ such that whenever $t \in U$, we have $\varrho(\sigma^t(f_j), f_j)<\varepsilon$ for $1 \leqq j \leqq n$ and $\varrho(\sigma^t(g_j^*), g_j^*)<\frac{\varepsilon}{3}$

for $1 \leqq j \leqq m$. Hence the triangular inequality for ϱ implies that for $t \in U$ we also have $\varrho(\sigma^t(g_j), g_j) < \varepsilon$ for $1 \leqq j \leqq m$. Since $\varepsilon > 0$ was arbitrary, this implies $A \cup B = \{f_1, \ldots, f_n, g_1, \ldots, g_m\}$ is jointly strongly recurrent.

For $k \geqq 1$ let P_k denote the set of all $f \in H$ which are obtained by analytic continuation of a Dirichlet series $\sum_{n=1}^{\infty} a_n n^{-z}$, $\operatorname{Re}(z) > 1$, where $\{a_n\}$ is a sequence of complex numbers which is periodic modulo k. We also let $P = \bigcup_{k=1}^{\infty} P_k$. Then we have:

4.11. THEOREM. *For any integer $k \geqq 1$, the following are equivalent:*
(a) *The Riemann hypothesis for $L(., \chi)$ for all Dirichlet characters χ modulo k.*
(b) *Each member of P_k is a strongly recurrent point of $(H, \mathbf{R}, \sigma)$.*

In consequence, the generalized Riemann hypothesis holds for all Dirichlet L-functions if and only if all points of P are strongly recurrent.

PROOF. Since for characters χ modulo k, $L(., \chi)$ is in P_k, (b)$\Rightarrow$(a) is a consequence of Theorem 3.7. To prove (a)$\Rightarrow$(b), let $f \in P_k$. Then we have $f(z) = \sum_{m=1}^{\infty} a_m m^{-z}$ for $\operatorname{Re}(z) > 1$, where $\{a_m\}$ is periodic modulo k. Let E_k be the set of all integers $m \geqq 1$ such that whenever a prime p divides m, p also divides k. Let $\chi_1, \ldots, \chi_n$ be the Dirichlet characters modulo k. For $1 \leqq j \leqq n$ and $m \in E_k$, let us define $\alpha_{m,j}$ by

$$\alpha_{m,j} = \frac{1}{n} \sum_{h=1}^{k} \overline{\chi_j(h)}\, a_{mh}, \quad 1 \leqq j \leqq n.$$

Let us now define g_j in H by

$$g_j(z) = \sum_{m \in E_k} \alpha_{m,j} m^{-z}, \quad z \in \Omega, \quad 1 \leqq j \leqq n.$$

Clearly the above series converges absolutely for $\operatorname{Re}(z) > 0$ and hence uniformly for z in Ω. Also, we have the formula

$$(*) \qquad f = \sum_{j=1}^{n} g_j L(., \chi_j)$$

which may be verified by direct manipulation for $\operatorname{Re}(z) > 1$ and hence by analytic continuation.

If (a) holds, then by Theorem 4.1, $\{L(., \chi_j)\colon 1 \leqq j \leqq n\}$ is jointly strongly recurrent. Hence, by Proposition 4.10, $\{L(., \chi_j)\colon 1 \leqq j \leqq n\} \cup \{g_j\colon 1 \leqq j \leqq n\}$ is jointly strongly recurrent. Hence by Proposition 2.6, f given by the formula $(*)$ is strongly recurrent. So we are done.

4.12. EXAMPLE. In view of Theorem 4.11 it might be natural to conjecture that whenever $f \in H$ is given by analytical continuation of a formula $f(z) = \sum_{n=1}^{\infty} a_n n^{-z}$, $\operatorname{Re}(z) > 1$, with $\{a_n\}$ bounded, f is a strongly recurrent point of $(H, \mathbf{R}, \sigma)$. The following example shows that this is false. Take $a_n = (-1)^{n-1} n^{-1/2}$, $n \geqq 1$. Then

$\{a_n\}$ is bounded, $f(z)=\sum_{n=1}^{\infty} a_n n^{-z}=(1-2^{3/4-z})\zeta\left(z+\frac{1}{4}\right)$. Thus f is entire and therefore may be regarded as a point in H. However this function f is not a strongly recurrent point of $(H, \mathbf{R}, \sigma)$. To prove this, let K be the compact subset of Ω given by

$$K=\left\{x+iy\colon \frac{5}{8}\leqq x\leqq\frac{7}{8},\ -\frac{\pi}{\log 2}\leqq y\leqq\frac{\pi}{\log 2}\right\}.$$

It is easy to see that f is nonzero on K. Therefore we may choose ε such that $0<\varepsilon<\operatorname{Inf}_{z\in K}|f(z)|$. If f were strongly recurrent, then the set A of all real t such that $\sup_{z\in K}|f(z+it)-f(z)|<\varepsilon$ would satisfy $\bar{\delta}(A)>0$, and in particular A would be unbounded. For $t\in A$, $K-it$ contains no zero of f. Since each point $\frac{3}{4}+\frac{2\pi mi}{\log 2}$ with $m\in Z$, $m\neq 0$, is a zero of f, this can not happen if $|t|$ is sufficiently large. This contradiction shows that f is not strongly recurrent.

5. On coherent flows

We begin with an inheritance theorem for coherence:

5.1. PROPOSITION. *Let (X, G, π) be a flow, $h\in G$, $h\neq 0$. Then (X, G, π) is coherent if and only if the discrete subflow (X, Z, π_h) is coherent.*

PROOF. This is an immediate consequence of Propositions 2.2 and 2.5.

Next we consider some negative examples. Let K be the space of all bisequences of signs. That is, K consists of sequences ε_n: $n\in Z$ with $\varepsilon_n=\pm 1$. We give it the topology of pointwise convergence. Clearly, there is a metric on K which induces the topology of pointwise convergence. Let π^1: $K\to K$ be the homeomorphism defined by $\pi^1(\varepsilon)=\varepsilon'$ when $\varepsilon'_n=\varepsilon_{n+1}$, $n\in Z$. π^1 induces a discrete flow (K, Z, π) on K. K is clearly a compact topological group under the operation of pointwise multiplication. Let μ be the unique Haar probability measure on K.

5.2. PROPOSITION. *The flow (K, Z, π) is not coherent. Indeed, whenever A is a Borel subset of K with $\mu(A)>0$, there exist ε, ε' in A such that both ε and ε' are strongly recurrent but $\{\varepsilon, \varepsilon'\}$ is not jointly strongly recurrent.*

PROOF. Relative to the probability measure μ, π^1 is a measure preserving and ergodic transformation (see [7, pp. 40 and 53]). Hence by the individual ergodic theorem ([7, p. 31]), we have, for every nonempty open set $U\subseteq K$,

$$\bar{d}(\{n\in Z\colon \pi^n(\varepsilon)\in U\})\geqq\mu(U)>0$$

holds for almost all (μ) ε in K.

Since the topology of K admits a countable basis, it follows that almost all (μ) points of K are strongly recurrent. That is, if L denotes the set of all strongly recurrent points of K, then $\mu(L)=1$. Therefore $\mu(A\cap L)=\mu(A)>0$. Hence, replacing A by $A\cap L$ if necessary, we may assume that $A\subseteq L$.

Suppose, now, that each pair ε, δ in A is jointly strongly recurrent. The map from K^2 to K sending (ε, δ) to $\varepsilon\delta^{-1}=\varepsilon\delta$ is a flow homomorphism. Hence by Proposition 2.6, the image $AA^{-1}=\{\varepsilon\delta^{-1}\colon \varepsilon\in A,\ \delta\in A\}$ of A under this map consists of strongly recurrent points. That is, we have $AA^{-1}\subseteq L$. But K is a compact group and A is a Borel subset with $\mu(A)>0$. Therefore Theorem B in [5, p. 68], suitably generalised, implies that AA^{-1} contains a neighbourhood of the identity 1 of K. Hence this neighbourhood is contained in L. That is, there exists an integer $N\geqq 1$ such that whenever ε in K satisfies $\varepsilon_n=1$ for $-N\leqq n\leqq N$, it follows that ε is strongly recurrent. This absurd conclusion establishes the proposition.

5.3. Proposition. *The flow* $(H, \mathbf{R}, \sigma)$ *is not coherent.*

Proof. Let $(K, \mathbf{Z}, \pi)$ be the flow introduced above. Let $\varphi\colon K\to H$ be defined by: for ε in K, $\varphi(\varepsilon)=f_\varepsilon$ where

$$f_\varepsilon(z)=\sum_{n=-\infty}^{\infty}\varepsilon_n(z-in)^{-2},\quad z\in\Omega.$$

Clearly φ is a flow homomorphism of $(K, \mathbf{Z}, \pi)$ into the discrete subflow modulo 1 $(H, \mathbf{Z}, \sigma_1)$ of $(H, \mathbf{R}, \sigma)$. Since φ a is 1—1 continuous map from the compact space K, it is a homeomorphism of K onto $\varphi(K)$. Thus φ embeds (K, Z, π) as a subspace restriction of (H, Z, σ_1). Hence Proposition 5.2 implies that (H, Z, σ_1) is not coherent. Hence by Proposition 5.1, $(H, \mathbf{R}, \sigma)$ is not coherent.

Next we proceed to obtain a sufficient condition for coherence. We begin with:

5.4. Definitions. We say that a flow (X, G, π) is *equicontinuous* in case for every x_0 in X and for every $\varepsilon>0$, there exists an $\eta>0$ (possibly depending on both x_0 and ε) such that whenever x in X satisfies $\varrho(x, x_0)<\eta$, we have for all t in G $\varrho(\pi^t(x), \pi^t(x_0))<\varepsilon$. That is, the requirement is that the family $\{\pi^t\colon t\in G\}$ should be equicontinuous at each point of X. Notice that equicontinuity of a flow depends only on the topology of the phase space X (and of course on the behaviour of the phase projection π) and not on the particular metric ϱ.

We say that a flow (X, G, π) is *locally equicontinuous* in case for every finite subset A of X, the subspace restriction $(\mathrm{Sp}\,(A), G, \pi)$ is equicontinuous.

Discrete flows induced by isometries of the phase space are clearly equicontinuous. The next proposition shows that these examples are typical.

5.5. Proposition. *Let* (X, G, π) *be a flow on a metric space* (X, ϱ). *Then* (X, G, π) *is equicontinuous if and only if there exists a metric* ϱ^* *on* X *which is equivalent to* ϱ *and relative to which each* π^t, $t\in G$, *is an isometry.*

Proof. The 'if' part is trivial. To prove the 'only if' part, we may assume that ϱ is bounded (replace, if necessary, ϱ by the equivalent metric $\min(1, \varrho)$). Define $\varrho^*\colon X\times X\to\mathbf{R}$ by $\varrho^*(x, y)=\sup\{\varrho(\pi^t(x), \pi^t(y))\colon t\in G\}$. A routine check shows that ϱ^* works.

5.6. Lemma. *Every equicontinuous flow is coherent.*

Proof. Since a discrete subflow of an equicontinuous flow is clearly equicontinuous, it is sufficient to consider (due to Proposition 5.1) discrete flows. Accordingly, let $(X, \mathbf{Z}, \pi)$ be a discrete flow which is equicontinuous. In view of Prop-

osition 5.5 we may assume that, relative to the given metric ϱ on X, π is an isometry of X.

Now let $\{x_1, \ldots, x_n\}$ be a finite subset of X such that each x_j, $1 \leqq j \leqq n$, is strongly recurrent. Let $\varepsilon > 0$. Then there exists $A_j \subseteq Z$ with $\bar{d}(A_j) > 0$ such that for $m \in A_j$, $\varrho(\pi^m(x_j), x_j) < \frac{\varepsilon}{2}$, $1 \leqq j \leqq n$. Hence if $r, s \in A_j$, then

$$\varrho(\pi^{r-s}(x_j), x_j) = \varrho(\pi^r(x_j), \pi^s(x_j)) \leqq \varrho(\pi^r(x_j), x_j) + \varrho(\pi^s(x_j), x_j) < \varepsilon.$$

Hence, if we put $B = \bigcap_{j=1}^{n} (A_j - A_j)$, where $A_j - A_j = \{r - s\colon r \in A_j,\ s \in A_j\}$, then for $m \in B$, we have $\varrho(\pi^m(x_j), x_j) < \varepsilon$ for $1 \leqq j \leqq n$. Since each A_j satisfies $\bar{d}(A_j) > 0$, Theorem 1 of [6] implies that $\bar{d}(B) > 0$. Hence $\{x_1, \ldots, x_n\}$ is jointly strongly recurrent, and we are done.

5.7 THEOREM. *Every locally equicontinuous flow is coherent.*

PROOF. Let (X, G, π) be a locally equicontinuous flow. Let A be a finite subset of X such that each point in A is strongly recurrent. Since by definition of local equicontinuity the subspace restriction $(\mathrm{Sp}\,(A), G, \pi)$ is equicontinuous, Lemma 5.6 implies that A is jointly strongly recurrent in $(\mathrm{Sp}\,(A), G, \pi)$. Hence A is jointly strongly recurrent in (X, G, π) and we are done.

Theorem 5.7 is clearly not the last word on coherence. The next proposition shows that locally equicontinuous flows exhibit a property which can not be expected of general coherent flows.

5.8. PROPOSITION. *A point in a locally equicontinuous flow is strongly recurrent if and only if it is almost periodic.*

PROOF. As before, it suffices to consider discrete flows. Since almost periodicity always implies strong recurrence, we only need to prove the 'only if' part. Accordingly, let $(X, \mathbf{Z}, \pi)$ be a discrete locally equicontinuous flow and let x_0 be a strongly recurrent point in this flow. Arguing, with the equicontinuous flows $(O(x_0), Z, \pi)$, as in the proof of Lemma 5.6, we see that for every $\varepsilon > 0$ there exists a set $A \subseteq Z$ with $\bar{d}(A) > 0$ such that whenever m is in $B = A - A$, we have $\varrho(\pi^m(x_0), x_0) < \varepsilon$. But by Theorem 2 of [6], B is syndetic. Hence x_0 is almost periodic.

Although the evidence is admittedly very scanty, the results of the previous section tempts one to make the following conjecture. Future work on coherence should be directed towards settling it.

5.9. CONJECTURE. Let D be the space of all $f \in H$ such that f is obtained by analytic continuation of a Dirichlet series $\sum_{n=1}^{\infty} a_n n^{-z}$, $\mathrm{Re}\,(z) > 1$, where a_n is a *bounded* sequence of complex numbers. Clearly D is an invariant subset of $(H, \mathbf{R}, \sigma)$. Therefore the subspace restriction $(D, \mathbf{R}, \sigma)$ is a flow in its own right. We *conjecture* that $(D, \mathbf{R}, \sigma)$ is coherent.

Since the flows $(D_k, \mathbf{R}, \sigma)$, $k \geqq 3$, of 4.6 are subspace restrictions of $(D, \mathbf{R}, \sigma)$, in view of Theorem 4.8 we have:

5.10. THEOREM. *The coherence conjecture in 5.9 implies the generalised Riemann hypothesis for all Dirichlet L-functions.*

Notice that Example 4.11 shows that all points of $(D, \mathbf{R}, \sigma)$ are not strongly recurrent.

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References

[1] B. Bagchi, A joint universality theorem for Dirichlet L-functions, *Math. Z.*, **181** (1982), 319—334.
[2] H. Bohr, Über eine quasi-periodische Eigenschaft Dirichletschen Reihen mit Anwendung auf die Dirichletschen L-Funktionen, *Math. Annalen*, **85** (1922), 115—122.
[3] W. H. Gottschalk and G. A. Hedlund, Recursive properties of topological transformation groups, *Bull. Amer. Math. Soc.*, **52** (1946), 488—489.
[4] W. H. Gottschalk and G. A. Hedlund, *Topological dynamics*, Amer. Math. Soc. Colloquium publications **36** (1955).
[5] P. R. Halmos, *Measure theory*, Van Nostrand (New York, 1950).
[6] C. L. Stewart and R. Tijdeman, On infinite difference sets, *Can. J. Math.*, **31** (1979), 897—910.
[7] P. Walters, *Ergodic theory — introductory lectures*, Springer Verlag (Berlin, 1975).

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